Document 3: Requirements Gathering

Cosmetics Store Analytics & Business Intelligence System

Document: 03 - Requirements Gathering **Team Name:** Team 5

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1. Stakeholder Analysis

Identifying stakeholders and analyzing their needs is essential for building an analytics system that provides practical business value. The primary stakeholders for this project and their core requirements are summarized below:

Stakeholder	Role in Store / Project	Key Needs & Expectations
Store Owner / General Manager	Oversees store operations and commercial profitability	Needs high-level summary reports on total revenue, gross profit, sales growth, and overall retail performance.
Inventory & Purchasing Manager	Manages product stock, supplier orders, and storage	Needs accurate stock visibility, identification of low-stock and negative-stock items, and product sales turnover rates.
Sales Staff & Cashiers	Executes daily customer checkout transactions	Needs consistent item pricing, accurate invoice totals, and reliable customer account classification.
Course Instructor & Evaluators	Academic review and assessment of the project	Needs clean, reproducible code, sound database normalization, structured documentation, and clear student defense.

2. User Stories & Use Cases

2.1 User Stories

Story 1 (Monthly Revenue Tracking): As a store manager, I want to review total monthly sales revenue and gross profit, so that I can evaluate business financial health and compare month-over-month performance.

Story 2 (Product Category Analysis): As an inventory manager, I want to see which product categories generate the highest sales volume, so that I can prioritize shelf space and order appropriate stock quantities.

Story 3 (Stock Discrepancy Auditing): As an inventory supervisor, I want to identify items with negative or zero stock quantities, so that the store team can conduct a physical inventory audit and correct data entry errors.

Story 4 (Customer Group Breakdown): As a business analyst, I want to separate walk-in retail sales from commercial account transactions, so that store management can understand revenue contributions from different client segments.

2.2 Representative Use Case: Generating a Sales Summary Report

- **Primary Actor:** Store Manager / Data Analyst.
- **Precondition:** Cleaned transaction records have been verified and loaded into the SQL database.
- **Main Flow of Events:**

1. The analyst or manager specifies the reporting period (e.g., specific month or quarter).
2. The system queries the database to retrieve relevant sales invoices and line items.
3. The system computes total revenue, total quantity sold, total cost, and gross profit margin.
4. The system groups metrics by product category and customer account type.
5. The system generates summary tables and visual trend charts for management review.

- **Postcondition:** The store manager reviews the verified summary report to support business decisions.

3. Functional Requirements

The system must provide the following functional capabilities, organized into four primary modules:

Module 1: Data Ingestion & Preprocessing

FR-1.1: The system shall import raw transaction files, item records, and account lists from standard CSV formats.

FR-1.2: The system shall detect and handle duplicate invoice records and missing attribute values.

FR-1.3: The system shall validate that line-item sums reconcile with invoice header totals within standard rounding tolerances.

Module 2: Relational Database Storage

FR-2.1: The system shall store transaction data in normalized SQL Server tables with primary keys and foreign key constraints.

FR-2.2: The system shall maintain referential integrity to prevent orphaned sales line items or invalid customer accounts.

FR-2.3: The system shall create appropriate database indexes to ensure efficient querying across date and category fields.

Module 3: Business Analytics & Calculations

FR-3.1: The system shall compute total revenue, units sold, and gross profit across customizable date ranges.

FR-3.2: The system shall aggregate sales metrics by product category and customer classification type.

FR-3.3: The system shall calculate positive inventory value, negative stock exposure, and net stock valuation.

Module 4: Visualization & Reporting

FR-4.1: The system shall produce visual charts showing monthly revenue trends and hourly transaction distribution.

FR-4.2: The system shall display department performance summaries and highlight top-selling product categories.

4. Non-Functional Requirements

To ensure reliability, usability, and academic defensibility, the system must satisfy the following non-functional requirements:

- **Usability:** The Python code, SQL queries, and project documentation must be written clearly using simple, consistent formatting so that university students and academic evaluators can easily understand and run every component.
- **Reliability & Data Integrity:** The data pipeline must produce consistent, deterministic results. All financial calculations and line-item aggregations must be mathematically reconciled with zero unexplained discrepancies.
- **Performance:** Data preprocessing scripts and database queries must run within reasonable execution times (under two minutes) on standard student laptop hardware without requiring specialized server infrastructure.
- **Security & Privacy:** Transaction datasets and database configurations must adhere to basic data privacy practices. No confidential customer contact details or database administrator passwords shall be hard-coded into public documentation.